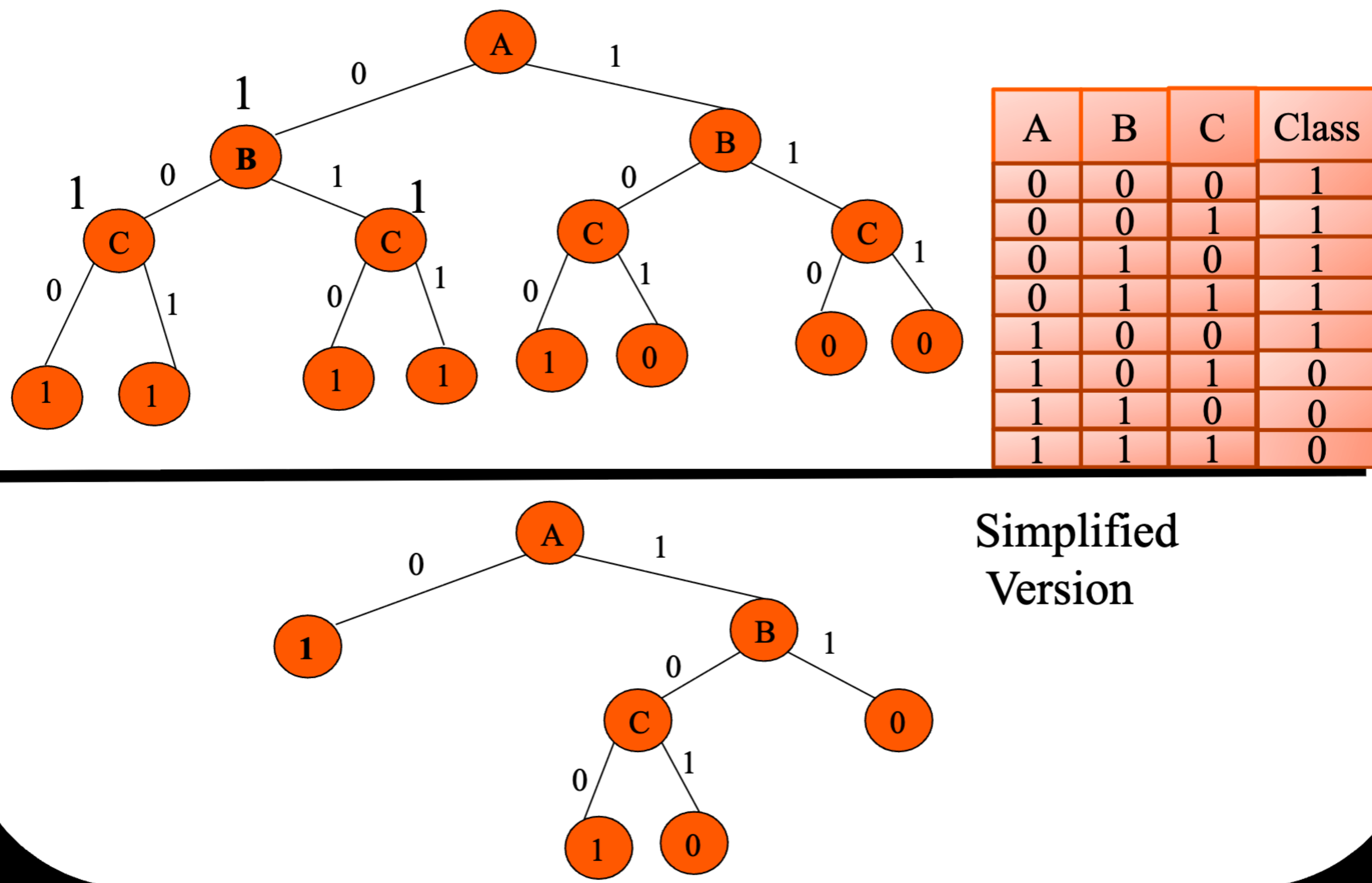


1) **Classification (Decision tree) Questions:** Draw the full decision tree for the parity function of four Boolean attributes, A, B, C , and D . Is it possible to simplify this tree? If yes, simplify it and redraw the simplified tree.

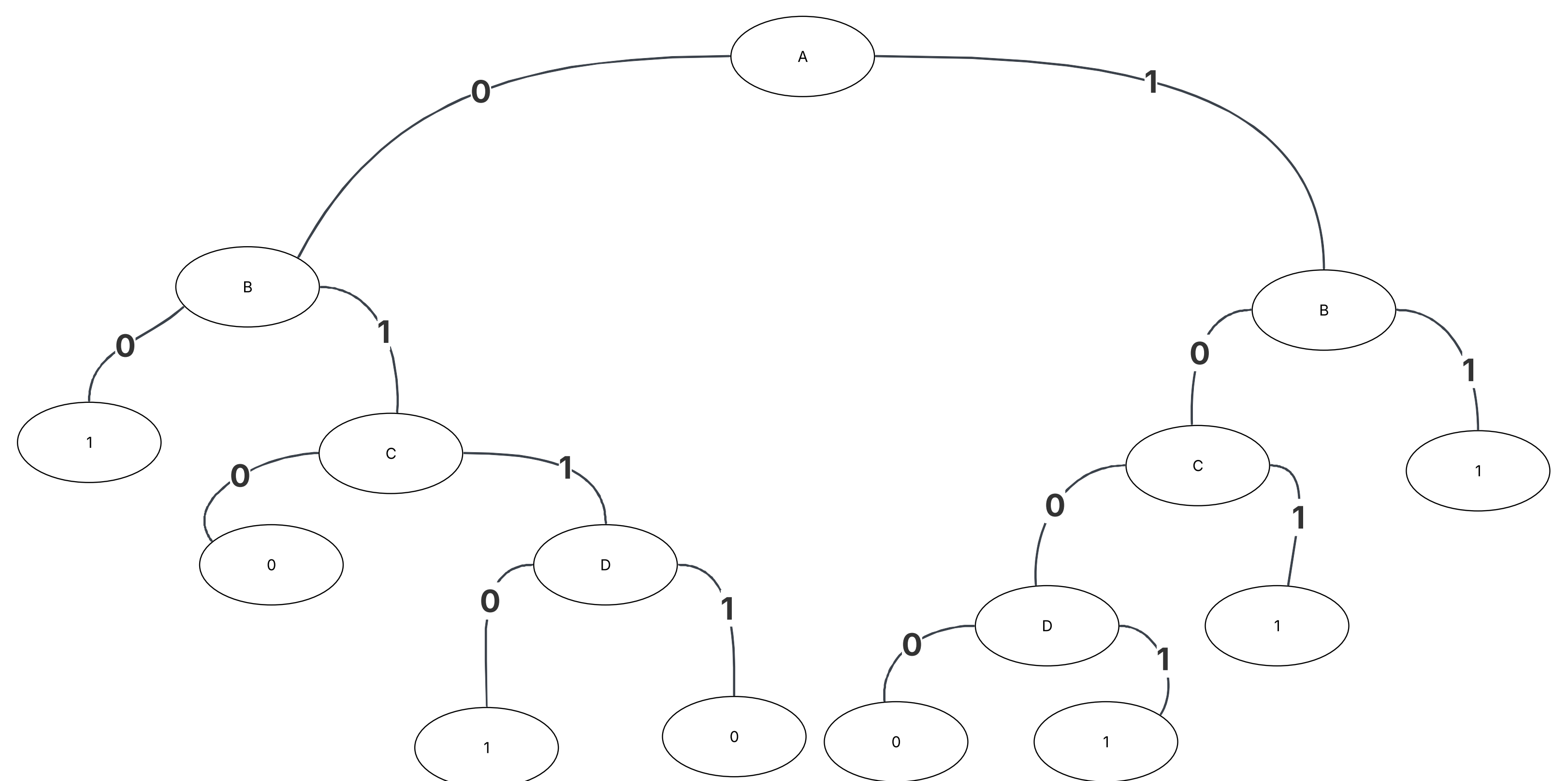
A	B	C	D	Class
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
1	0	1	1	1
1	0	0	0	0
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

- **Example the Decision tree**

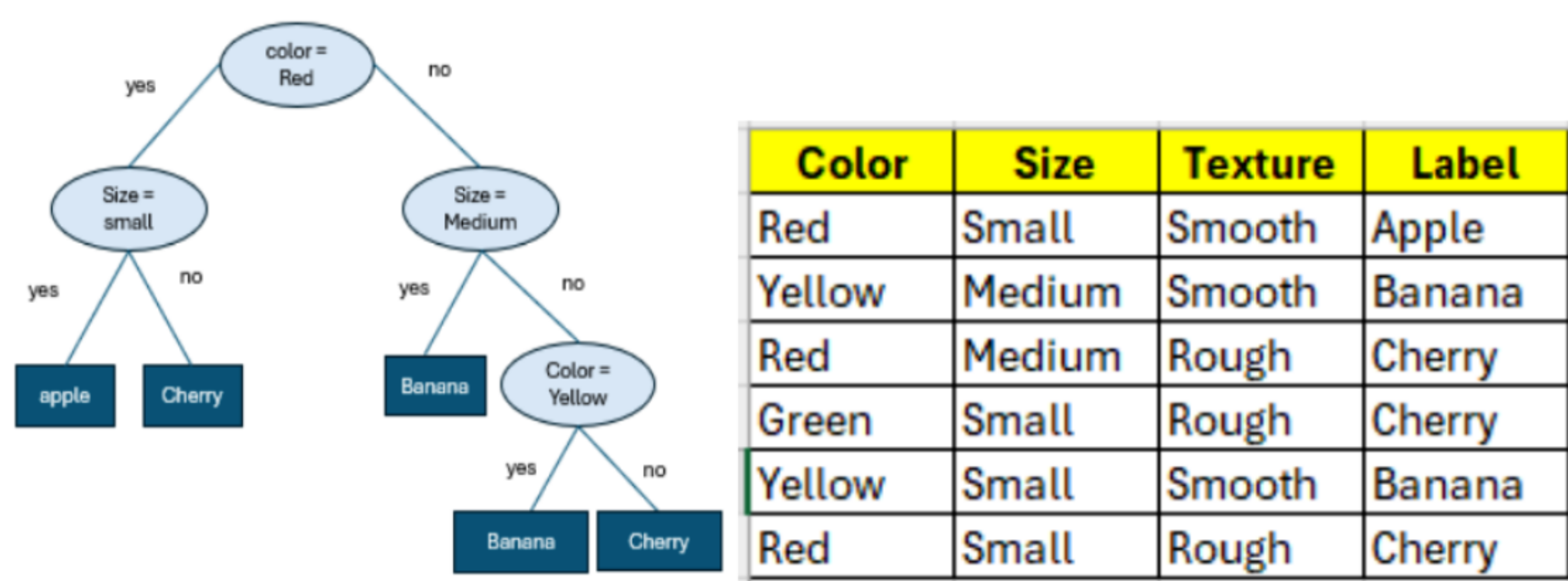


2) **Classification (Decision tree) Question:** Draw the full decision tree for the parity function of four Boolean attributes, A , B , C , and D . Is it possible to simplify this tree? If yes, simplify it and redraw the simplified tree

A	B	C	D	Class
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	0
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1



3) Based on the following decision tree model and the given data find out if the model overfits, underfits, or generalizes well.



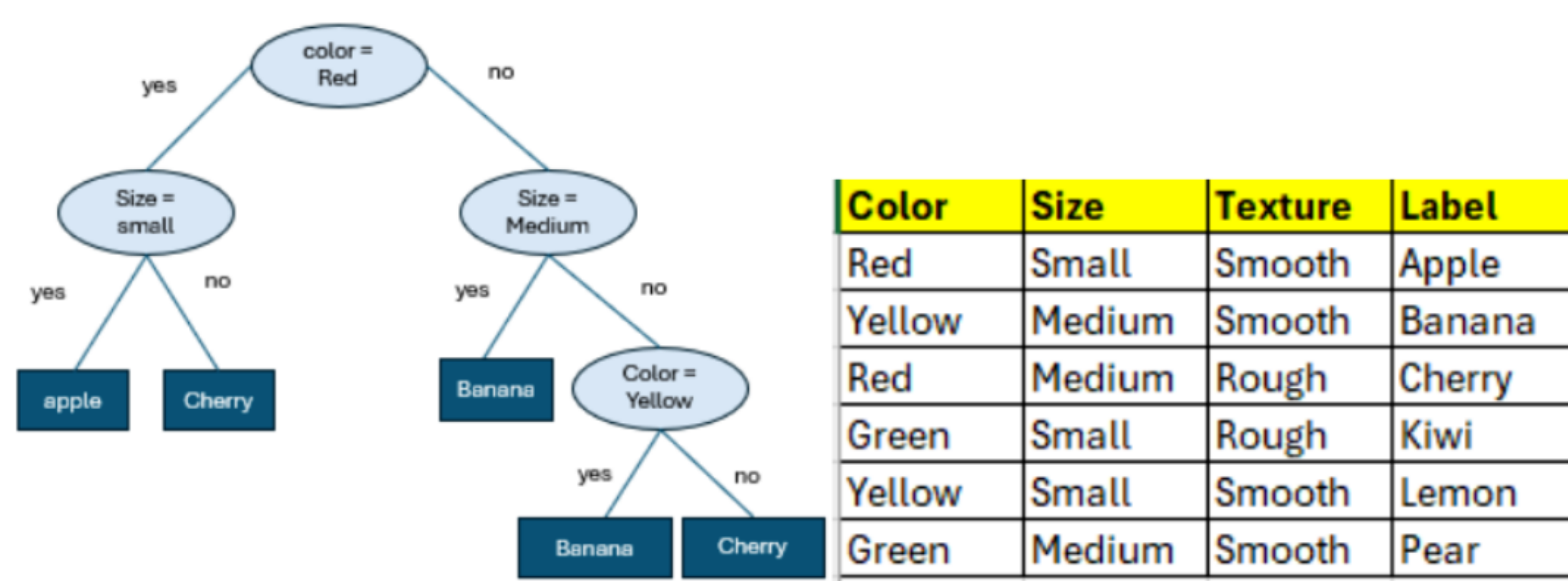
This decision tree model is an example of generalization because it correctly identifies 5/6 of the labels in the dataset.

4) Based on the following decision tree model and the given data find out if the model overfits, underfits, or generalizes well.



This decision tree model is an example of Underfitting, model is too simple.

5) Based on the following decision tree model and the given data find out if the model overfits, underfits, or generalizes well.



This decision tree model is an example of Underfitting, it fails to classify several fruits (same tree as #3 but it is unable to classify unforeseen data)